

Smart Fall-Detection Bracelet for Independent Elderly Safety

Project title: Development of a Smart Fall Detection Bracelet for Elderly Safety

Team Info: Section M, Tutorial #5, Team #3

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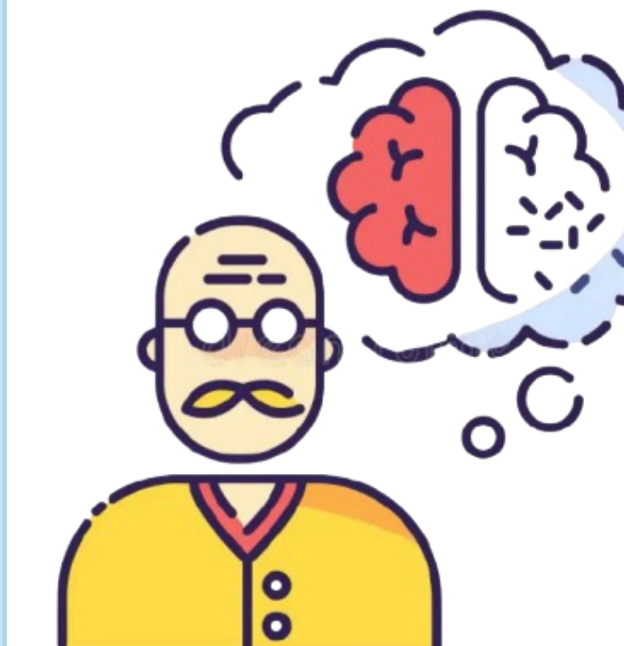
The Problem and Foundation

Problem Statement

Final Problem Statement

- Elderly individuals struggle to find fall-safety alert devices that are affordable and reliably responsive in real-world situations. Fear of falling and remaining unnoticed for long periods causes many seniors to limit daily activities, increasing physical and emotional distress. Current solutions, such as wearable alert devices and emergency buttons, often detect only certain types of falls or require user interaction to request help, leaving unresponsive seniors without timely assistance.

User Research Insights:

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- Fear of falling and being left unattended
 - Often avoid using devices that feel uncomfortable
 - Strong need for automatic detection
 - Concern about delayed emergency response
 - Prefer simple, easy-to-use devices

Requirements:

- Accurate fall detection using motion/health sensors
- Automatic alert system (no user confirmation required)
- Comfortable for daily, long-term wear
- Minimal false alarms



Constraints:

- Cost < \$50
- Limited size and battery capacity
- Must be simple and easy to use for elderly users



Design Journey

Design Process Overview

Problem → Ideation → Concept Generation → Selection

Dimension	Weight	★ Concept A	Concept B	Concept C
Feasibility	0.38	1.52	1.9	0.76
Desirability	0.26	1.04	0.78	0.52
Responsibility	0.1	0.5	0.4	0.3
Viability	0.24	1.2	0.72	0.24
Creativity	0.02	0.02	0.06	0.1
Total	1	4.28	3.86	1.92

Selection reasoning: Concept A scored highest overall and best balanced across all the criteria. Its wrist worn factor directly addresses user comfort concerns, and existing sensor technology makes it feasible within the \$50 budget constraint.

Final Design Concept = Smart Bracelet



Description: Wrist-worn fall-detection bracelet with an accelerometer that uses motion and heart-rate sensors to detect sudden movements and abnormal physiological changes. When a fall is detected, the device automatically sends an emergency alert to a caregiver.

Testing & Value

Prototype & Testing

What was built:

A prototype bracelet with sensor-based fall detection using an accelerometer and heart-rate monitoring.

Why:

To evaluate comfort, usability, and the effectiveness of automated fall detection.

Results:

- The bracelet form factor was comfortable, familiar, and easy to wear.
- Sensor-based detection is feasible for identifying falls.
- Automatic alerts improve response time and safety.

Iteration Highlight

VERSION 1

Motion detection only

- Basic concept focusing only on accelerometer based motion detection.
- High false alarm rate from everyday movements.

VERSION 2

Added heart-rate function

- Heart-rate monitoring + improved alert logic.
- More reliable detection with significantly reduced false alarms and faster response.

Key Idea from iteration:

Combining motion AND physiological data produces meaningfully better results than either sensor alone.

VALUE PROPOSITION

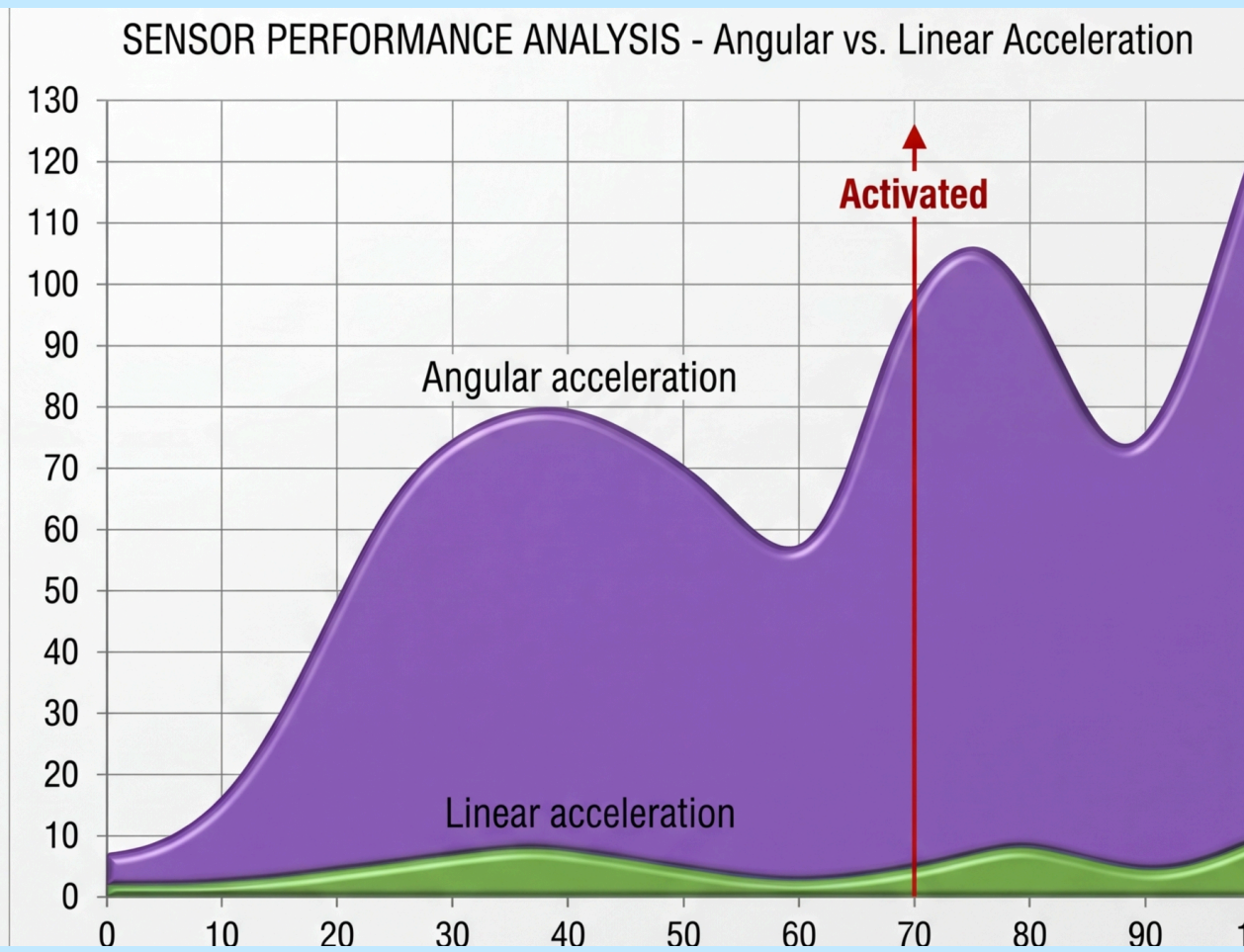
User Pains Relieved

- Fear of falling alone
- Delayed emergency response
- Uncomfortable devices
- Unreliable (false alarms)
- Hard-to-use devices

User Gains Created

- Faster emergency assistance
- Improved safety
- Greater independence
- Comfortable daily wear
- Reduced caregiver anxiety

Reflection & Next Steps



Key Learning

- If a device is uncomfortable, users won't wear it.
- Criteria must reflect real user needs, not just engineering convenience.

Unresolved Need:

Ensuring high detection accuracy while minimizing false alarms remains a challenge.

PATH FORWARD

Fully functional prototype:

develop a complete working device with refined sensors to reduce false positives.

Real-user testing: conduct structured trials with elderly participants for feedback on comfort and accuracy.

FUTURE ENHANCEMENTS

- Improve detection accuracy with trained ML models on fall data.
- Optimize battery life for multi-day wear.
- Explore additional health monitoring (blood oxygen, irregular heart rhythm).
- Design caregiver mobile app for alert management.